THE EXTENSION BY ZERO ALONG A QUASI-COMPACT OPEN IMMERSION COMMUTES WITH LIMITS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file proetale.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the reduction to the commutation with products, the expression of the extension by zero as the kernel of the displayed short exact sequence, and the comparison of the commutation with products for the three functors involved using the exactness of products.

Editorial note. The label lemma-j-shriek-limits names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source proetale.tex –label lemma-j-shriek-limits –start 5401 –end 5428 –local-id 28002 returned RESULT=OK (L28002 free, occupancy FREE, public interval clear). Candidate taken from sources/stacks-proetale/candidates-halfA.json.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- The short exact sequence associated to a quasi-compact open immersion. Same chapter, the lemma giving the short exact sequence 0 -> j_!G -> j_*G -> i_*i^{-1}j_*G -> 0 for the inclusion of the complement with its reduced structure. Supplies the exact sequence expressing the extension by zero as a kernel.
- Commutation with products of the pushforwards and of the pullback along a closed immersion. Same chapter, the lemma that the pullback along a closed immersion commutes with products. Supplies the commutation with products of i^{-1}, while j_* and i_* commute with products as right adjoints.
- Exactness of products in the category of sheaves on the pro-etale site. Cohomology on Sites, the proposition on enough weakly contractibles, which implies that products are exact in the category of abelian sheaves. Supplies the exactness of products, which passes the commutation with products from the other terms of the sequence to j_!.

Source reference. Original reference 01HO: sources/stacks-proetale/pinned-original-source.tex, source lines 5162–5162. Prerequisite pointer invoked at proof line(s) 5423; it names a standard fact of the same chapter that the counted argument

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uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 095L: sources/stacks-proetale/pinned-original-source.tex, source lines 5352–5352. Prerequisite pointer invoked at proof line(s) 5414; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition 0947: sources/stacks-chow/tags.txt, source lines 11540–11540. Prerequisite pointer invoked at proof line(s) 5427; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $j : U \to X$ be a quasi-compact open immersion morphism of schemes. The functor $j_! : Ab(U_{pro\text{-}étale}) \to Ab(X_{pro\text{-}étale})$ commutes with limits.*

Proof. Since $j_!$ is exact it suffices to show that $j_!$ commutes with products. The question is local on X, hence we may assume X affine. Let $\mathcal{G}$ be an abelian sheaf on $U_{pro\text{-}étale}$. We have $j^{-1}j_*\mathcal{G} = \mathcal{G}$. Hence applying the exact sequence of Lemma 095L we get

$$0 \to j_!\mathcal{G} \to j_*\mathcal{G} \to i_*i^{-1}j_*\mathcal{G} \to 0$$

where $i : Z \to X$ is the inclusion of the reduced induced scheme structure on the complement $Z = X \setminus U$. The functors j_* and i_* commute with products as right adjoints. The functor i^{-1} commutes with products by Lemma 01HO. Hence $j_!$ does because on the pro-étale site products are exact (Cohomology on Sites, Proposition 0947). □